

**A PROPOSED OFFERING OF A PROMOTIONAL WITH
LOYALTY MEMBERSHIP AND FEEDBACK SYSTEM FOR
ALEGRIA'S BAKEHOUSE**

A Project Proposal Presented to the

Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

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INTRODUCTION

The introduction includes the background information about the system, the problem or the need for the system, and the main objective of this project.

Background Information

Alegria's Bakehouse is a local bakery that provides freshly baked breads, pastries, cakes, and beverages to customers in the community. Many people visit the bakehouse regularly for daily food needs, special occasions, and simple treats. As the number of customers continues to grow, it becomes important for the business to maintain good customer relationships and encourage repeat visits through effective promotions and engagement.

At present times, Alegria's Bakehouse uses basic promotional methods such as in-store announcements and social media postings. While these methods help inform customers about products and offers, they do not fully support customer loyalty and long-term engagement. There is no organized way to register loyal customers, monitor promotional participation, or gather customer feedback. Because of this, customer preferences and satisfaction levels are not properly recorded, which limits the ability of the business to improve its services and promotions.

To address these concerns, this project proposes a Promotional with Loyalty Membership and Feedback System for Alegria's Bakehouse. The system will serve as an online website platform where customers can view current promotions, register as loyalty members, earn rewards, and submit feedback based on their experience. This website will also allow the business to manage promotions, monitor loyalty membership activities, and review customer feedback in a more organized and efficient manner.

The proposed system aims to improve customer engagement by providing a simple and accessible way for customers to interact with the business. Loyalty membership features encourage repeat purchases, while the feedback section allows customers to share

their opinions and suggestions. Through this, Alegria's Bakehouse can better understand customer needs and identify areas for service improvement.

The main objective of this project is to support the growth of Alegria's Bakehouse by improving promotional management, strengthening customer loyalty, and enhancing customer satisfaction. By using a Promotional with Loyalty Membership and Feedback System business can build stronger relationships with its customers, improve decision-making, and maintain consistent service quality. Overall, the project aims to help Alegria's Bakehouse become more customer-focused and competitive through the use of a user-friendly website.

CLIENT INFORMATION

This chapter provides the details about the client organization, business background and their relevant information.

Client Organization

Business Name: Alegria's Bakehouse

Client Name: Ma. Sienne Joy B. Diamante & Gerald Ken V. Cajida

Contact Information

Address: 115 B MacArthur Highway, Marulas, Valenzuela

Phone: +63 927 474 9403

Email: alegriasbakehouse.ph@gmail.com

Relevant Images:



Figure 1. Alegria's Bakehouse Logo



Figure 2. Alegria's Bakehouse Owners



Figure 3. Client Establishment

Business and Industry

Operating in the food and beverage industry, Alegria's Bakehouse is a local bakery business that produces and sells freshly baked breads, pastries, cakes, and beverages. The business serves walk-in customers and depends on consistent product quality and customer satisfaction to maintain regular customers. As part of its daily operations, Alegria's

Bakehouse manages promotions and interacts with repeat customers using manual methods. This setup makes it difficult to properly track loyal customers, organize promotional activities, and collect customer feedbacks. To support the business, the proposed Promotional with Loyalty Membership and Feedback System aims to improve customer engagement, reduce manual processes, and help the management make informed decisions while supporting the continued growth of the bakery.

PROJECT SCOPE

This chapter includes the deliverables, inclusion, exclusions, assumptions, and constraints that could affect the outcome of the proposed system.

Deliverables

The project will deliver a fully functional Promotional with Loyalty Membership and Feedback System for Alegria's Bakehouse, which includes a customer module that enables users to view current promotions and announcements, register as loyalty members, earn and track loyalty points, redeem rewards when applicable, and submit feedback and ratings. The system will also include an admin or management module that allows staff to manage promotions and announcements, view and maintain loyalty member records, monitor loyalty points and rewards, and review and analyze customer feedback. A centralized database will be developed to store customer information, promotions, loyalty points, and feedback data. In addition, complete system documentation such as the system requirements specification, user manual, and technical documentation will be provided, along with the final project report and system presentation.

Inclusion

The scope of the project includes an online system that is accessible through both desktop and mobile browsers, customer registration and loyalty membership management, promotion posting and management by the administrator, a feedback and rating system for customers, secure login for administrators, and basic system reports such as the number of registered loyalty members and customer feedback summaries.

Exclusion

The project does not include online ordering or delivery services, online payment processing, point-of-sale (POS) system integration, mobile application development for Android or iOS platforms, advanced data analytics or AI-based recommendations, inventory and sales management systems, or SMS marketing automation features.

Assumptions

It is assumed that the client will provide accurate business information and timely feedback during system development, that users have basic knowledge of using an online platform, that stable internet access is available for both administrators and customers, and that the loyalty program rules are clearly defined and approved by the client prior to implementation.

Constraints

The project is constrained by the availability of the client for reviews, feedback, and approvals, the limited budget and resources typical of a student-developed system, and the limited development time imposed by the academic schedule.

PROJECT APPROACH

This section explains how the project will be carried out from start to finish. It describes the steps, methods, and activities that will be used to develop the project and ensure that the goals are achieved effectively and on time.

Overview of the Proposed Approach

The project will follow a structured and iterative development approach to ensure that the system meets its objectives and user requirements. The process begins with a thorough analysis of the problem and stakeholder needs, followed by careful planning, design, development, testing, and deployment. Continuous feedback and evaluation will be incorporated to improve system quality, usability, and performance throughout the project lifecycle. Methodologies and Frameworks to be used

The project will primarily use the Agile methodology, allowing flexibility, gradual development, and continuous improvement. Agile is suitable for this project as it enables regular testing, early problem identification, and adaptation to changing requirements. Key Agile practices such as sprint planning, iterative development, and regular reviews will be applied. Supporting tools and frameworks for documentation, version control, and testing will also be utilized to ensure efficient collaboration and system reliability.

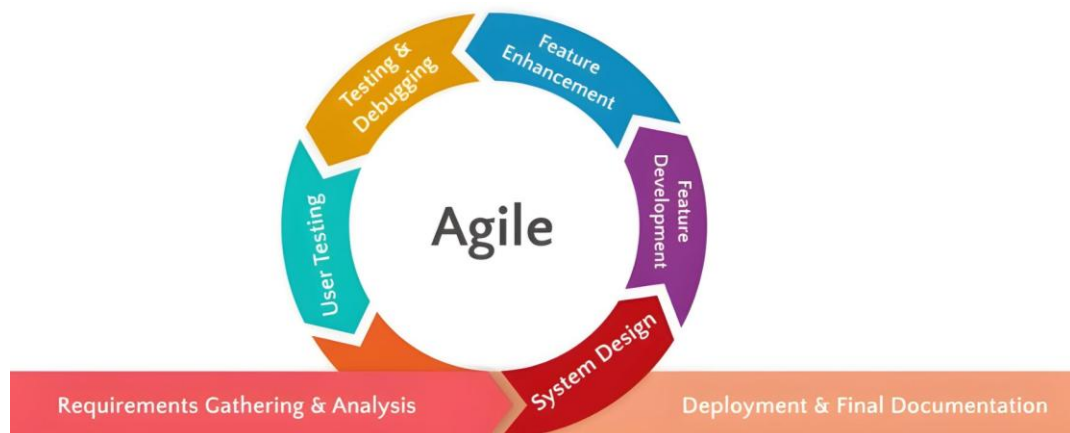


Figure 4. Agile Methodology Diagram

Key Activities and Milestones

Key activities include requirements gathering and analysis, system design, development of core features, testing and debugging, system evaluation, and final deployment. Major milestones of the project are the completion of the project proposal, approval of system requirements, completion of the system prototype, final system implementation, user testing and evaluation, and submission of the final documentation and project presentation.

PROJECT TEAM

The project will be developed by a group of nine members working throughout the planning, development, and implementation phases. The team is committed to delivering a functional and user-friendly Promotional with Loyalty Membership and Feedback System by combining each member's skills, experiences, and contributions.

	Project Leader
 Mikyla Jade F. Galacete	Prior experience in team leadership and project management. Has knowledge in system development that helps in planning and understanding the system. Responsible for leading the team, organizing tasks, analyzing client requirements, and ensuring the system aligns with user needs.
	Back-end Developer
 Jayson V. Bernante	I have prior experience working on thesis projects and specialize in back-end development tasks such as managing databases, implementing server-side processes, and maintaining overall system functionality. I am adaptable to various responsibilities based on the team's needs to help ensure the project is completed successfully. I am also committed to continuously developing my technical skills in both back-end and front-end areas.

	Front-end Developer
 Angel Patulot	I have experience as a thesis programmer. I can be flexible depending on what the group needs to be able to finish the project. I am very interested in learning both back-end and front-end development.
	UI/UX / Front-end Developer
 Elizalde Yordan	Responsible for the UI/UX design of this project. Has knowledge in web/system development, especially in front-end development. I focus on creating clean, user-friendly, and visually appealing interfaces to improve the overall user experience and overall project standards.
	UI/UX Designer
 Maria Lucia Ampado	Has experience serving as a project leader and has demonstrated strong leadership and organizational skills. She is also capable in UI/UX design and contributes actively to creating user-friendly and visually effective interfaces

	System Analyst / Quality Assurance
 Ben O. Onde	Has experience in system analysis, requirement evaluation, software testing, and usability assessment. Responsible for analyzing client requirements, reviewing system processes, and assisting in the design validation of the system.
	System Tester and Documentation
 Lea C. Camallere	Has experience as Project Leader, handling project documentation by compiling research, technical references, and user guides. She also assists in system testing to verify that the implemented features align with documented requirements and overall project standards.
	System Analyst
 Allen Christian Earl Huerto	Manages both system logic and data-related processes. He connects frontend functions with backend operations, oversees data storage, and evaluates user interaction data to help improve system features.

	Quality Assurance and Documentation
 Sean Roscoe Panday	Responsible for maintaining system reliability and performance. He conducts thorough testing, prepares test cases, and identifies system issues to ensure that all features such as loyalty point tracking function accurately and consistently.

Table 1. Project Team for Promotional with Loyalty Membership and Feedback System

PROJECT TIMELINE

The project timeline only marks the key milestones, the anticipated outcomes, and the dependencies of each stage of this project.

Month	Sprint	Deliverables and Milestones
Month 1	Sprint 1 – Project initiation	Requirements gathering, stakeholder interviews, Initial Product Backlog creation.
Month 2	Sprint 2 – System Analysis	Defining data requirements, System Architecture & Dataflow
Month 3	Sprint 3 – UI/UX Creation	Creating wireframes for landing page, Branding alignment, Style guide
Month 4	Sprint 4 – Prototyping	Developing interactive mock-ups, Clickable UI/UX Prototype
Month 5	Sprint 5 – Design Development & Proposal	Finalizing the mobile responsive layout, Approved Technical Proposal
Month 6	Sprint 6 – Tech Stack Setup	Installation of Frameworks, Development Environment Setup
Month 7	Vacation Phase – Core Backend Logic	(ACAD BREAK) Coding of database schema. Initial Backend Setup, Working Database, Login Framework
Month 8	Sprint 7 – Membership Module Build	Development of Member Registration forms, Functional Membership Modul
Month 9	Sprint 8 – Loyalty Engine Development	Loyalty Points Engine Completion

Month 10 - 11	Sprint 9 – Feedback and Interaction Module	Costumer Engagement Features
Month 12	Sprint 10 – Admin Dashboard & Analytics	Develoment of the management portal for Algeria’s Bakehouse, Management Analytics Dashboard
Month 13	Sprint 11 – System Integration & Hardening	Full-scale integration of all modules, Feature Complete System (BETA)
Month 14	Sprint 12 – QA & UAT	Cross-browser and mobile responsiveness testing, internal bug hunting and resolution. Bug-Free System Report.
Month 15	Sprint 13 – Development & Handover	Final system development hosting, Finalization of Documentation, Final Project Defense and Official Handover of the System to the Client.

Table 2. Project Timeline for Promotional with Loyalty Membership and Feedback System

PROJECT RESOURCES

This section identifies the essential resources including hardware, software, and client-provided resources necessary for the successful development and implementation of the Promotional with Loyalty Membership and Feedback System.

Hardware

- **Laptops/Desktops:** with at least 8GB RAM and Intel i5 (or equivalent) processors for efficient coding and multitasking.
- **Processor (CPU):** Intel Core i3 (11th gen up)
- **Memory RAM:** 8 GB
- **Storage:** 256 GB SSD
- **Operating System:** Windows 10 / Windows 11
- **Mobile Devices:** Android and iOS smartphones for testing responsive web design and mobile browser compatibility.
- **Network:** High-speed internet connection for cloud database synchronization and version control.

Minimum requirement to host a system using mobile device:

- **Processor:** Octa-core 2.0 GHz (Snapdragon 6 series / Helio G80 or equivalent)
- **RAM:** 4 GB
- **Storage:** 64 GB (Mainly for browser cache and system logs)
- **Display:** 6.1-inch LCD/OLED (To test UI responsiveness and readability)
- **Operating System:** Android 10 up or iOS 14 (and above)
- **Browser:** Updated Google Chrome, Safari, or Microsoft Edge
- **User Experience (UX):** Testing the ease of navigation for loyalty members.
- **Member Access:** Simulating how a customer will register, track points, and submit feedback on-site at the bakehouse.

Software

- **Visual Studio Code:** This is the development tools that we will use and to input the codes. HTML5, CSS, JAVASCRIPT, PHP, and BOOTSTRAP.
- **Database:** MySQL to store loyalty points, member records, and feedback data.
- **Design Tools:** The design tools that use to create UI/UX wireframing is Figma.
- **Documentation:** Microsoft Office or Google Docs for the System Requirements Specification, User Manuals, Documentation, and Project Presentation.

Client-Provided Resources

As per the project assumptions, the client (Alegría's Bakehouse) shall provide:

- **Brand Assets:** High-resolution logos and shop images.
- **Content:** Promotional details, reward tier structures, and menu information.
- **Feedback:** Timely review of prototypes during the Agile sprints

Budget Allocation

The budget allocation tells how the domain name server and the database hosting will be function with a lowcost pricing. Below is a budget allocation and strategy for hosting the "Promotional with Loyalty Membership and Feedback System":

1. Primary Hosting Strategy: Low-Cost/Free Tier

To ensure the system remains accessible after the developer's graduate, the project recommends selecting low-cost or free-tier hosting services.

- **Recommended Platforms:** They can use Firebase or Github pages to be able to run and the best candidate to use for hosting.
- **Estimated Cost:** No cost for initial deployment, provided usage remains within the provider's free limits.

2. Database Hosting

The system requires a centralized database to store customer records, loyalty points, and feedback.

- MySQL or PostgreSQL.
- They can utilize cloud-based database services that offer a forever free or limited free tier (such as **PlanetScale** for MySQL, **Supabase** for PostgreSQL, or the integrated database options within firebase without costing).

3. Domain Name

While the proposal focuses on free services, a professional web address www.alegriasbakehouse.com would require a separate purchase. Domain Name service can cost P500 pesos to P800 pesos.

4. Sustainability and Maintenance Resources

Rather than a high financial budget for hosting, the project allocates resources in the form of documentation to ensure long term viability:

- We will provide a manual instruction to help the owner to manage the system.
- We will create a video tutorial for them to watch on how update the menu prices and to add/remove the images and the prices.
- To address potential connectivity issues at the Valenzuela branch, the project includes a manual backup protocol. A physical Loyalty Ledger will be maintained to record transactions during internet outages, ensuring that service remains uninterrupted and data is preserved for later synchronization.

RISK MANAGEMENT

This Risk Management plan identifies potential obstacles that could affect the development or daily operation of the system at Alegria's Bakehouse. Because the bakery currently relies on manual methods to track customers and promotions through social media, transitioning to a digital platform introduces specific technical, operational, and legal risks. The goal of this risk management plan is to ensure the system remains reliable, secure, and beneficial to the business long after its initial deployment.

Risk	Description	Possible Impact	Mitigation Strategy
Loyalty Point Exploitation	Customers might attempt to "game" the system by submitting multiple fake feedback entries or creating duplicate accounts to earn free rewards/products	Financial loss for the bakehouse due to unearned rewards; skewed data for management decision-making	Implement "Rate Limiting" on feedback submissions (example, one per purchase); require a unique transaction ID or receipt number to validate loyalty point additions.
Connectivity & Store Access	Potential for unstable internet connectivity at the physical Valenzuela branch, preventing the website from loading during a customer transaction.	Frustrated customers who cannot see their rewards; loss of trust in the "digital" loyalty program.	Maintain a physical 'Loyalty Backup Ledger' at the counter. If the internet is unavailable, staff will manually record the customer's registered phone number and the transaction amount. These entries will

			then be manually encoded into the system by the admin once the connection is stable.
Privacy Compliance (DPA 2012)	Improper handling of customer names, contact numbers, and emails collected during loyalty registration.	Potential legal issues regarding the Philippines' Data Privacy Act; loss of business reputation for Alegria's Bakehouse	Include a clear "Privacy Consent" checkbox on the registration page; ensure only authorized student developers and the bakery owner have access to the customer database.
Sustainability & Handover	As a student project, the system may lack technical support or hosting funds once the developers graduate from Datamex.	The website becomes inaccessible or "broken" within six months, forcing the bakery back to manual methods.	Select low-cost or free-tier hosting services (like Infinityfree or GitHub Pages); provide a "Maintenance Manual" to the owner and record a video tutorial on how to update menu prices and promotions.

Table 3. Risk Management for Promotional with Loyalty Membership and Feedback System

COMMUNICATION PLAN

The communication plan for the proposed system is designed to ensure effective coordination between the development team, the client, and the end-users throughout the entire project lifecycle. The system serves as a promotional platform integrated with a loyalty membership and feedback system, aiming to improve customer engagement, satisfaction, and retention.

At the initial phase of the project, communication will focus on client interviews and requirement gathering. The development team will conduct structured interviews with the client to understand existing business processes, promotional strategies, loyalty program mechanisms, and feedback management practices. This phase is essential in identifying user requirements, system objectives, and functional specifications. All information gathered during the interviews will be documented and reviewed to ensure accuracy and alignment with the client's expectations.

The team will proceed with analysis and internal coordination. Meetings will be conducted to discuss the gathered requirements and translate them into system features, including promotional announcements, customer registration, loyalty point tracking, rewards management, and feedback submission. Clear communication among team members will help define roles, responsibilities, and development timelines.

Once the requirements are finalized, the project will move to the UI/UX design phase. Wireframes will be created and presented to the client for evaluation. Feedback from the client will be used to refine the interface to ensure usability, accessibility, and consistency with the organization's branding. This phase ensures that the system design meets both functional and aesthetic requirements before development begins.

During the system development phase, continuous communication will be maintained through progress reports and scheduled updates. The client will be informed about development milestones, system functionalities, and any issues encountered. Any necessary changes or clarifications in system requirements will be communicated promptly to avoid delays and misinterpretations.

Following development, communication will focus on system testing and validation. The client and selected users will participate in testing activities to evaluate the system's functionality, usability, and performance. Feedback collected during this phase will be analyzed and used to implement necessary revisions and improvements prior to system deployment.

Communication will extend to user orientation and support. The development team will provide system documentation, user guides, and demonstrations to ensure proper system usage. Post-implementation communication will also be conducted to collect user feedback and assess the effectiveness of the promotional features, loyalty membership system, and feedback mechanism for future enhancements.

PROJECT GOVERNANCE

The Promotional with loyalty membership and feedback system for Alegria's Bakehouse will follow a structured governance framework to ensure that all development activities align with the goals and expectations of the client. This approach promotes accountability, transparency, and efficient decision-making throughout the project lifecycle.

Roles and Responsibilities

Roles	Responsibilities
Client Representative	Responsible for submitting project requirements, approving significant modifications, and receiving regular progress updates.
Project Leader	Manages the entire project, ensures effective communication among all stakeholders, and maintains high quality standards.
Back - End developer	deals with the hidden processes that run behind the scenes.
Front - End developer	builds the visible, interactive parts of websites and web applications using languages like HTML, CSS, and JavaScript, ensuring a good user experience across devices.
System Analyst	Analyzes client needs, aids in creating system specifications, and ensures alignment with business objectives.

Data Gatherer	Responsible for gathering accurate data from various sources, conducting interviews or surveys, and ensuring the integrity of the information for analysis and reporting.
UI/UX Designer	Responsible for creating a user-friendly interface.
Documentation	Basically, keeping on track and recording the progress that can be served as record or proof.
QA	to ensure a product or service meets specified requirements and standards.
System Tester	Responsible for conducting thorough testing to ensure the system's functionality and performance.

Table 4. Project Governance for Promotional with Loyalty Membership and Feedback System

APPROVAL

This Project Proposal entitled "**PROMOTIONAL WITH LOYALTY MEMBERSHIP AND FEEDBACK SYSTEM FOR ALEGRIA'S BAKEHOUSE**", prepared and submitted by **MARIA LUCIA AMPADO, JAYSON V. BERNANTE, LEA C. CAMALLERE, MIKYLA JADE F. GALACETE, ALLEN CHRISTIAN EARL HUEARTO, BEN O. ONDE, SEAN ROSCOE D. PANDAY, ANGEL PATULOT, and ELIZALDE YORDAN JR.** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, is hereby recommended for oral examination and approval.

Approved By:

Mr. Gabriel Thomas Torneros

Capstone Project Adviser

Mrs. Ma. Sienne Joy B. Diamante

Owner, Alegria's Bakehouse

Mr. Gerald Ken V. Cadija

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